

Rakesh R J

Associate Software Engineer · .NET Developer | Computer Vision Engineer

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 Portfolio

PROFESSIONAL SUMMARY

Software engineer with ~2.7 years of total professional experience spanning .NET development, Azure cloud services, REST API integration, SQL data operations, and computer vision engineering. Seeking opportunities to leverage enterprise application development, cloud, and AI skills in a challenging and growth-oriented environment while delivering impactful solutions.

WORK EXPERIENCE

Accenture

Jan 2025 – Present | ~1.9 Years

Associate Software Engineer (ASE)

► Project: Roche Diagnostics — Production Application & Data Support

Enterprise production support involving .NET, Azure, SQL, and REST API-based data workflows.

- Handled and resolved production tickets related to application, data, and processing issues.
- Analyzed tickets and investigated root causes to identify the required resolution.
- Performed database changes and data updates using **SQL** based on business requirements.
- Used **REST APIs and Swagger/OpenAPI** to trigger and process data in the production environment.
- Uploaded and managed required data using **Azure Storage Accounts**.
- Triggered downstream **data processing workflows/pipelines** through Azure.
- Monitored and validated data processing results after API and Azure operations.
- Troubleshoot issues involving .NET applications, databases, APIs, and data-processing workflows.
- Followed production **change and incident-management processes** while working with live systems.
- Collaborated with the team to resolve production tickets and complete requested changes efficiently.

Technologies: .NET | C# | SQL | Azure | Azure Storage | REST API | Swagger/OpenAPI | Azure Data Pipelines | Production Support

MacV

Apr 2024 – Nov 2024 | ~8 Months (Full-Time + Internship)

Computer Vision Engineer

► Project 1: PPE & Safety Alerting System

AI-based safety monitoring system for PPE violations and forklift-related incidents.

- Developed Python-based components for processing computer vision detection results.
- Used **gRPC** for inter-service communication between microservices.
- Built alert-processing workflows for PPE and forklift use cases.
- Used **MongoDB** to manage alert configurations, jobs, channels, and related data.
- Integrated **AWS SQS** for asynchronous alert processing and **AWS Lambda**-based notification workflows.
- Integrated notification services including **AWS SES/SNS** for real-time alerting.

Technologies: Python | gRPC | MongoDB | AWS SQS | AWS Lambda | AWS SES | AWS SNS

► Project 2: Gemini AI Video Analysis & Alerting

AI-powered video analysis using Gemini for safety event detection (helmet violations, falls, harness).

- Developed and modified **FastAPI** backend services for video ingestion and processing.
- Integrated **Gemini AI** for video analysis and generated structured detection results.
- Created and maintained **prompts** for detection scenarios (helmet violations, falls, safety harness).
- Processed videos stored in **Amazon S3** and optimized frame sampling rates for performance.
- Integrated detection results with an alerting pipeline via **AWS SQS** and stored data in **MongoDB**.

Technologies: Python | FastAPI | Gemini AI | Prompt Engineering | AWS S3 | AWS SQS | MongoDB | REST API

► Project 3: Vehicle Detection & Number Plate Recognition

Computer vision system for vehicle detection, tracking, classification, and ANPR.

- Implemented vehicle detection using **YOLOv8** with object tracking across video frames.
- Performed number plate detection and text extraction using **EasyOCR**.
- Classified vehicles into categories: cars, buses, trucks, bikes, ambulances, and auto-rickshaws.
- Implemented vehicle color classification and generated structured output (Track ID, type, plate, confidence, color).
- Optimized video processing using **frame skipping and frame-similarity** techniques.

Technologies: Python | YOLOv8 | OpenCV | EasyOCR | TensorFlow/Keras | Object Tracking | Computer Vision

TECHNICAL SKILLS

Programming & Data	Python, SQL, C# / .NET, Java, C
.NET Development	.NET, C#, ASP.NET, Production Support, Troubleshooting, Debugging
Cloud & DevOps	Microsoft Azure, Azure Storage Accounts, Azure Data Pipelines, Git / GitHub, CI/CD fundamentals
APIs & Integration	REST APIs, Swagger / OpenAPI, API Testing, Data Integration
AI & Computer Vision	Python, OpenCV, YOLOv8, EasyOCR, FastAPI, Gemini AI, TensorFlow, Keras, CNN, Object Tracking, Prompt Engineering
Cloud (AWS)	AWS SQS, AWS Lambda, AWS SES, AWS SNS, Amazon S3
Databases & Tools	SQL, MongoDB, gRPC, VS Code, IntelliJ, Jupyter Notebook, Power BI, Git
Web	HTML/CSS, JavaScript, React.js, PHP

EDUCATION

Kongu Engineering College	Apr 2024
B.E. Computer Science Engineering — CGPA: 8.45 / 10.0	
Veveaham Hr. Sec. School	Mar 2020
HSC (Class XII) — 70% SSLC (Class X) — 84.8%	